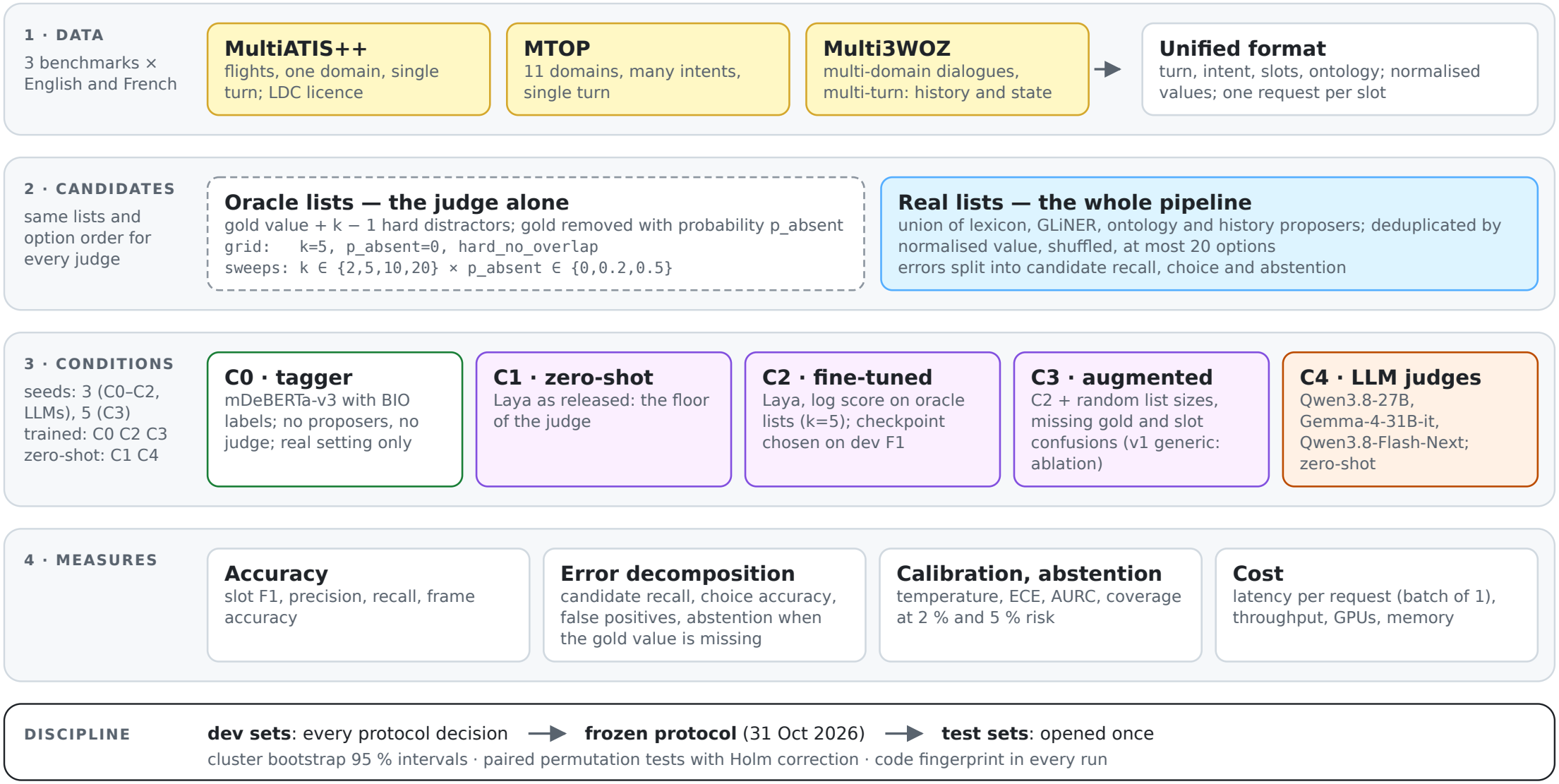
1 · DATA
3 benchmarks × English and French
MultiATIS++
flights, one domain, single turn; LDC licence
MTOP
11 domains, many intents, single turn
Multi3WOZ
multi-domain dialogues, multi-turn: history and state
Unified format
turn, intent, slots, ontology; normalised values; one request per slot
2 · CANDIDATES
same lists and option order for every judge
Oracle lists — the judge alone
gold value + k − 1 hard distractors; gold removed with probability p_absent
grid: k=5, p_absent=0, hard_no_overlap
sweeps: k ∈ {2,5,10,20} × p_absent ∈ {0,0.2,0.5}
Real lists — the whole pipeline
union of lexicon, GLiNER, ontology and history proposers; deduplicated by normalised value, shuffled, at most 20 options
errors split into candidate recall, choice and abstention
3 · CONDITIONS
seeds: 3 (C0–C2, LLMs), 5 (C3)
trained: C0 C2 C3
zero-shot: C1 C4
C0 · tagger
mDeBERTa-v3 with BIO labels; no proposers, no judge; real setting only
C1 · zero-shot
Laya as released: the floor of the judge
C2 · fine-tuned
Laya, log score on oracle lists (k=5); checkpoint chosen on dev F1
C3 · augmented
C2 + random list sizes, missing gold and slot confusions (v1 generic: ablation)
C4 · LLM judges
Qwen3.8-27B, Gemma-4-31B-it, Qwen3.8-Flash-Next; zero-shot
4 · MEASURES
Accuracy
slot F1, precision, recall, frame accuracy
Error decomposition
candidate recall, choice accuracy, false positives, abstention when the gold value is missing
Calibration, abstention
temperature, ECE, AURC, coverage at 2 % and 5 % risk
Cost
latency per request (batch of 1), throughput, GPUs, memory
DISCIPLINE
dev sets: every protocol decision → frozen protocol (31 Oct 2026) → test sets: opened once
cluster bootstrap 95 % intervals · paired permutation tests with Holm correction · code fingerprint in every run